MILITARY SPECIFICATION

GYROSWITCH, RATE: M46

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers special requirements not included on the drawings and provides the quality assurance provisions for the fabrication of parts, assembling, and preparation for delivery of one type of switching device designated as Gyroswitch, Rate M46 used to sense missile roll rate and act as a ground safing device, with the Lance warhead section.

1.2 WARNING - The M46 gyroswitch is a potentially dangerous device. It is pressurized to gas pressures of 1650 psi maximum, and if accidentially initiated by rupture of the frangible seal, injury may occur to personnel.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-48528(PA) - Piston Motor Assembly.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-331 - Fuze and Fuze Components Environmental and Performance Tests For
MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8877827 - Cap Assembly, Preload.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

8877874 - Terminal Assembly.
8877919 - Rotor Assembly.
8877928 - Cap, Exhaust.
8877941 - Gimbal Assembly.
8878743 - Dome Subassembly.
8878784 - Wafer Assembly, Cage.
8878793 - Bellows Assembly, Cage.
8878813 - Bellows Assembly, Exhaust.
8878817 - Dome Assembly.
9978819 - Gyroswitch, Rate M46.
8878821 - Ring Assembly, Contact.
8878826 - Terminal Assembly.
8878844 - Data Plate.
8878847 - Connector Assembly.
8878857 - Piston Motor Assembly.
9274158 - Container, Metal Ammunition, For Gyroswitch M46.
9285346 - Support Assembly, Gimbal.
9285347 - Mount Assembly, Center.
9285348 - M46 Gyroswitch Unwelded Assembly.
ET-8878819 - Equipment Tabulation.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The gyroswitch shall comply with all requirements specified on the drawing (dwg) 8878819, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-N-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Case leakage. The gyroswitch shall have an indicated leakage rate of helium tracer gas of less than 9×10^{-9} ATM cubic centimeters per second (cc/sec), (air equivalent) as specified in 4.4.1.

3.7 Circuit resistance, (non-operating). The circuit resistance between terminals listed in Table I shall not exceed the respective ohmic levels when tested as specified in 4.4.2.

Table I

J10 Terminals	Resistance (ohms)	Test Code (see 3.5)
a to E	0.25 max	061001
$\overline{b}$ to M	0.25 max	061002
$\overline{B}$ to V	open circuit	061003
B to W	open circuit	061004
B to Y	open circuit	061005
B to Z	open circuit	061006
S to N	*1.35 max	061007
C to S	0.25 max	061008
T to D	*1.35 max	061009
T to R	0.25 max	061010
U to J	**460 $\pm$ 90	061011

*Initiation circuit - Limit current to 0.02 amp max.
**Pickoff coil - Limit current to 0.015 ampere maximum.

3.8 Current carrying capacity. The maximum allowable voltage drop between terminals listed in Table II shall not exceed the respective levels when tested as specified in 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

Table II

J10 Terminals	Maximum Allowable Voltage Drop (Volts)	Test Code (see 3.5)
a to E	0.125	071001
b to M	0.125	071002
T to R	1.25	071003
S to C	1.25	071004

Note: Do not apply current to the following terminals: C to N, D to R, U to J, T to D, and S to N.

3.9 Insulation resistance. The insulation resistance between each circuit and all other circuits (see Table III) and case connected together as one circuit and grounded, shall be not less than 20 megohms when tested as specified in 4.4.4. Unused terminals shall be connected together and tested as one terminal.

Note: Terminals N to C and D to R shall be shorted before proceeding with this test. Terminals U to J shall not be tested.

Table III

J10 - a, b, V, W, Y, Z, B, C, R

J10 - Unused terminals - A, F, H, K, L, P, X, c, G

3.10 Functional operation. The pressurized gyroswitch, while being functioned as specified in 4.4.5 shall comply with the requirements described in 3.10.1 through 3.10.2.

3.10.1 "No-Go" operation (low input rate: 380°/Second). The gyroswitch shall comply with the requirements tabulated in Table IV at the specified temperatures. In all cases, after the input rate of 380°/second has been attained, the gyroswitch shall be initiated as specified in 4.4.5.3. For tests in Cycles 2 and 3, the gyroswitch shall be preconditioned in accordance with 4.4.6.2 and 4.4.6.3 respectively and test completed within 5 minutes after removal from the temperature chamber.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

Table IV

Cycle #1 @ + 80° ± 20°F

Test Parameters	J10 Terminals	Acceptance Criteria	Test Code (See 3.5)
a. Function Switch Closure	Y - B	No closure	091001
	Z - B	No closure	091002
b. Time to uncage from initiation	a - E	0.5 sec max	091003
	$\overline{M}$ - b	0.5 sec max	091004
	T - $\overline{R}$	0.5 sec max	091005
	S - C	0.5 sec max	091006
	B - V	0.5 sec max	091007
	W - B	0.5 sec max	091008
c. Total time unit remains uncaged	a - E	10 sec min	091009
	$\overline{M}$ - b	10 sec min	091010
	T - $\overline{R}$	10 sec min	091011
	S - C	10 sec min	091012
	B - V	10 sec min	091013
	W - B	10 sec min	091014
d. Cage switch Voltage Drop	V - B	0.125 volts (max)	091015
	W - B	0.125 volts (max)	091016
**e. Rotor spin rate at 0.5 sec (rpm) after initiation	U - J		091017
**f. Max rotor spin rate (rpm)	U - J		091018
**g. Rotor spin rate 10 sec after initiation (rpm)	U - J		091019
**h. Gyroswitch gas charging pressure	N/A	1100 to 1400 psig at 72°F ± 3°F	091020
(1) **i. Gyroswitch gas charge weight	N/A		091021

(1) Not applicable when operated by approved simulation of piston motor frangible seal activation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

Table IV (Con't)

Cycle #2 @ + 165°F ± 5°F

a.	Function Switch Closure	Y - B	No closure	091022
		Z - B	No closure	091023
b.	Time to uncage from initiation	a - E	0.5 sec max	091024
		M - b	0.5 sec max	091025
		T - R	0.5 sec max	091026
		S - C	0.5 sec max	091027
		B - V	0.5 sec max	091028
		W - B	0.5 sec max	091029
c.	Total time unit remains uncaged	a - E	10 sec min	091030
		M - b	10 sec min	091031
		T - R	10 sec min	091032
		S - C	10 sec min	091033
		B - V	10 sec min	091034
		W - B	10 sec min	091035
d.	Cage switch	V - B	0.125 volts (max)	091036
		W - B	0.125 volts (max)	091037
**e.	Rotor spin rate at 0.5 sec (rpm)	U - J		091038
**f.	Max rotor spin rate (rpm)	U - J		091039
**g.	Rotor spin rate 10 sec after initiation (rpm)	U - J		091040
**h.	Gyroswitch charging pressure	N/A	1100 to 1400 psig @ 72°F ± 3°F	091041
(1)**i.	Gyroswitch gas charge weight	N/A		091042

(1)Not applicable when operated by approved simulation of piston motor frangible seal activation.

Cycle #3 @ -40°F ± 5°F

a.	Function Switch Closure	Y - B	No closure	091043
		Z - B	No closure	091044

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

Table IV (Con't)

b. Time to uncage from initiation	a - E	0.5 sec max	091045
	M - b	0.5 sec max	091046
	T - R	0.5 sec max	091047
	S - C	0.5 sec max	091048
	B - V	0.5 sec max	091049
	W - B	0.5 sec max	091050
*c. Total time unit remains uncaged	a - E	10 sec min	091051
	M - b	10 sec min	091052
	T - R	10 sec min	091053
	S - C	10 sec min	091054
	B - V	10 sec min	091055
	W - B	10 sec min	091056
d. Cage Switch Voltage Drop	V - B	0.125 volts (max)	091057
	W - B	0.125 volts (max)	091059
**e. Rotor spin rate at 0.5 sec (rpm) after initiation	U - J		091059
**f. Max rotor spin rate (rpm)	U - J		091060
**g. Rotor spin rate 10 sec after initiation (rpm)	U - J		091061
**h. Gyroswitch charging pressure	N/A	1100 to 1400 psig @ 72°F ± 3°F	091062
(I)**i. Gyroswitch gas charge weight	N/A		091063

**Record values obtained.

(I)Not applicable when operated by approved simulation of piston motor frangible seal activation.

*Continuously monitor for the first 5 seconds ± 0.5 seconds. Thereafter, the gyroswitch shall comply with requirement d which shall be concluded prior to, but not equal to, the 10 second requirement for test parameter c. After test d, test parameter c shall be again monitored to assure compliance with its acceptance criteria.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:56Z
Check the source to verify that this is the current version before use.

3.10.2 "Go" operation.

3.10.2.1 "Go" operation (input rate: 1200°/second). The gyroswitch shall comply with the requirements set forth in Table V. After the input rate of 1200°/second has been attained, the gyroswitch shall be initiated as prescribed in 4.4.5.3.

Table V

Cycle @ 80°F ± 20°F

Test Parameters	J10 Terminals	Acceptance Criteria	Test Code (See 3.5)
a. Time to function switch closure from initiation	Y - B	0.6 sec max	092101
	Z - B	0.6 sec max	092102
*b. Total time function switches remain closed	Y - B	10 sec max	092103
	Z - B	10 sec max	092104
c. Time to uncage from initiation	V - B	0.5 sec max	092105
	W - B	0.5 sec max	092106
	a - E	0.5 sec max	092107
	b - M	0.5 sec max	092108
	T - R	0.5 sec max	092109
	S - C	0.5 sec max	092110
d. Total time unit remains uncaged	V - B	10 sec min	092111
	W - B	10 sec min	092112
	a - E	10 sec min	092113
	b - M	10 sec min	092114
	T - R	10 sec min	092115
	S - C	10 sec min	092116
e. Function and cage switch voltage drop	Y - B	0.08 volts (max)	092117
	Z - B	0.08 volts (max)	092118
	V - B	0.125 volts	092119
	W - B	0.125 volts (max)	092120

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

Table V (Con't)

**f.	Rotor spin rate 0.5 sec after initiation (rpm)	U - J		092121
**g.	Max rotor spin rate	U - J		092122
**h.	Rotor spin rate 10 sec after initiation (rpm)	U - J		092123
**i.	Gyroswitch charging pressure	N/A	1100 to 1400 psig at 72°F ± 3°F	092124
(1)**j.	Gyroswitch gas charge weight	N/A		092125

(1)Not applicable when operated by approved simulation of piston motor frangible seal activation.

*Continuously monitor for the first 5 seconds ± 0.5 seconds. Thereafter, the gyroswitch shall comply with requirement e which shall be concluded prior to, but not equal to, the 10 seconds requirement for test parameter b. After test e, test parameter b shall be again monitored to assure compliance with its acceptance criteria.

**Record values obtained.

3.10.2.2 "Go" operation (input rate: 500°/second). The gyroswitch shall comply with the requirements tabulated in Table VI at the specified temperatures. In all cases, after the input rate of 500°/second has been attained, the gyroswitch shall be initiated as specified in 4.4.5.3. For tests in cycles 2 and 3. the gyroswitch shall be preconditioned in accordance with 4.4.6.2 and 4.4.5.3 respectively and the test shall be completed within 5 minutes after removal from the temperature chamber.

Table VI

Cycle #1 @ 80°F ± 20°F

Test Parameters	J10 Terminals	Acceptance Criteria	Test Code (See 3.5)
a. Time to uncage from initiation	V - B	0.5 sec max	092201
	W - B	0.5 sec max	092202
	a - E	0.5 sec max	092203
	$\overline{b}$ - M	0.5 sec max	092204
	$\overline{T}$ - R	0.5 sec max	092205
	S - C	0.5 sec max	092206

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

Table VI (Con't)

b.	Time to function switch closure from initiation	Y - B Z - B	0.6 sec max 0.6 sec max	092207 092208
c.	Function and cage switch voltage drop	Y - B	0.08 volts (max)	092210
		Z - B	0.08 volts (max)	092212
		V - B	0.125 volts (max)	092209
		W - B	0.125 volts (max)	092211
d.	*Total time function switches remain closed	Y - B Z - B	10 sec min 10 sec min	092213 092214
e.	Total time unit remains uncaged	V - B W - B a - E b - M T - R S - C	10 sec min 10 sec min 10 sec min 10 sec min 10 sec min 10 sec min	092215 092216 092217 092218 092219 092220
**f.	Rotor spin rate 0.5 sec after initiation (rpm)	U - J		092221
**g.	Max rotor spin rate (rpm)	U - J		092222
**h.	Rotor spin rate 10.0 sec after initiation (rpm)	U - J		092223
**i.	Gyroswitch charging pressure	N/A	1100 to 1400 psig at 72°F ± 3°F	092224
(1)**j.	Gyroswitch gas charge weight	N/A		092225

(1)Not applicable when operated by approved simulation of piston motor frangible seal activation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

Cycle #2 @ 165°F ± 5°F

Table VI (Con't)

a.	Time to uncage from initiation	V - B	0.5 sec max	092226
		W - B	0.5 sec max	092227
		a - E	0.5 sec max	092228
		b - M	0.5 sec max	092229
		T - R	0.5 sec max	092230
		S - C	0.5 sec max	092231
b.	Time to function switch closure from initiation	Y - B	0.6 sec max	092232
		Z - B	0.6 sec max	092233
c.	Function and cage switch voltage drop	Y - B	0.08 volts (max)	092234
		Z - B	0.08 volts (max)	092237
		V - B	0.125 volts (max)	092234
		W - B	0.125 volts (max)	092236
d.	*Total time function switches remain closed	Y - B	10 sec min	092238
		Z - B	10 sec min	092239
e.	Total time unit remains uncaged	V - B	10 sec min	092240
		W - B	10 sec min	092241
		a - E	10 sec min	092242
		b - M	10 sec min	092243
		T - R	10 sec min	092244
		S - C	10 sec min	092245
**f.	Rotor spin rate 0.5 sec after initiation (rpm)	U - J		092246
**g.	Max rotor spin rate (rpm)	U - J		092247
**h.	Rotor spin rate 10.0 sec after initiation (rpm)	U - J		092248
**i.	Gyroswitch charging pressure	N/A	1100 to 1400 psig @ 72 °F ± 3°F	092249

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

Table VI (Con't)

(1)**j.	Gyroswitch gas charge weight	N/A		092250

(1)Not applicable when operated by approved simulation of piston motor frangible seal activation.

Cycle #3 @ - 40°F ± 5°F

a.	Time to uncage initiation	V - B	0.5 sec max	092251
		W - B	0.5 sec max	092252
		a - E	0.5 sec max	092253
		b - M	0.5 sec max	092254
		T - R	0.5 sec max	092255
		S - C	0.5 sec max	092256
b.	Time to function switch closure	Y - B	0.6 sec max	092257
		Z - B	0.6 sec max	092258
c.	Function and cage switch voltage drop	Y - B	0.08 volts (max)	092260
		Z - B	0.08 volts (max)	092262
		V - B	0.125 volts (max)	092259
		W - B	0.125 volts (max)	092261
d.	*Total time function switches remain closed	Y - B	10 sec min	092263
		Z - B	10 sec min	092264
e.	Total time unit remains uncaged	V - B	10 sec min	092265
		W - B	10 sec min	092266
		a - E	10 sec min	092267
		b - M	10 sec min	092268
		T - R	10 sec min	092269
		S - C	10 sec min	092270
**f.	Rotor spin rate 0.5 sec after initiation (rpm)	U - J		092271
**g.	Max rotor spin rate (rpm)	U - J		092272

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

Table VI (Con't)

**h.	Rotor spin rate 10.0 sec after initiation (rpm)	U - J		092273
**i.	Gyroswitch charging pressure	N/A	1100 to 1400 psig @ 72°F ± 3°F	092274
(1)**j.	Gyroswitch gas charge weight	N/A		092275

*Continuously monitor for the first 5 seconds ± 0.5 seconds. Thereafter, the gyroswitch shall comply with requirement c, which shall be concluded prior to, but not equal to, the 10 seconds requirement for test parameter d. After test c, test parameter d shall be again monitored to assure compliance with its acceptance criteria.

**Record values obtained.

(1)Not applicable when operated by approved simulation of piston motor frangible seal activation.

3.11 Environmental testing (see 4.4.6).

3.11.1 Transportation vibration. With the gyroswitch preconditioned to the specified temperatures, it shall withstand vibrations along each of its three mutually perpendicular axes. During this test, the gyroswitch shall be maintained at the specific temperature requirements. This test shall be conducted as specified in 4.4.6.1.

3.11.1.1 Transportation vibration @ 165°F + 5°F. After the gyroswitch has been temperature conditioned as specified in 4.4.6.2 and while maintained at the prescribed temperature, it shall be tested to the vibration spectrum defined in Table VII.

3.11.1.2 Transportation vibration @ - 65°F +5°F. After the gyroswitch has been temperature conditioned as specified in 4.4.6.4 and while maintained at the prescribed temperature, it shall be tested to the vibration spectrum defined in Table VII.

Table VII

Frequency Range Hertz (Hz)	Sweep Rate	Acceleration (g), Double Amplitude (da)
5-15	Frequency shall be cycle logarithmically from 5 Hz to 500 Hz to 5 Hz in 15 minutes	0.4 g
15-75		0.036" da
75-500		10.0 g

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

3.11.2 Impact shock. The gyroswitch shall withstand shocks of 50 gravity units (g) ± 5 g in both directions along each of its three mutually perpendicular axes (see Figure 1) for a total of 6 shocks. Each shock shall last for 11 milliseconds (ms) + 1 ms in duration. Before the gyroswitch is shocked, it shall be preconditioned to the specified temperature requirements. This test shall be conducted as specified in 4.4.6.5.

3.11.2.1 Impact shock @ 165 ± 5°F. After the gyroswitch has been conditioned as specified in 4.4.6.2, it shall be shocked within 5 minutes after removal from the temperature chamber.

3.11.2.2 Impact shock @ -65 ± 5°F. After the gyroswitch has been conditioned as specified in 4.4.6.4, it shall be shocked within 5 minutes after removal from the temperature chamber.

3.11.3 High temperature @ 165 ± 5°F. The gyroswitch shall be capable of withstanding exposure to a temperature of 165°F when tested as specified in 4.4.6.2.

3.11.4 Low temperature @ -65 ± 5°F. The gyroswitch shall be capable of withstanding exposure to a temperature of -65°F when tested as specified in 4.4.6.4.

3.11.5 Temperature shock. The gyroswitch shall withstand sudden changes in temperature when tested as specified in 4.4.6.6.

3.11.6 Temperature and humidity. The gyroswitch shall be capable of withstanding exposure to a temperature-humidity cycle as specified in 4.4.6.7.

3.12 Gas charge pressure and weight. Gas charge pressure measured at the specified temperature shall comply with dwg 8878819. Gas charge pressure shall be measured in accordance with 4.4.7. Every gyroswitch before and after pressurization shall be weighed so as to give the gas charge weight (see 4.4.7).

3.13 Soldering. All applicable soldering shall comply with the requirements specified in MIL-STD-1460.

3.14 Components. Objective evidence (see 6.2) shall be made available to the Government representative that the Piston Motor Assembly Conforms to MIL-P-48528(PA), and drawing (8878857).

3.15 Flatness of base. Flatness of base of the gyroswitch shall be determined as in 4.4.8. Temperature of the gyroswitch at which optical flatness measurement is made shall be recorded to the nearest one degree and shall be within the temperature range from 69 to 75°F. The acceptance limits of flatness shall be as shown in Table VIII.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

Table VIII

Optical measurement of base flatness

Temperature, °F	Flatness Range in Light Bands	
	Concave (Minus)	Convex (Plus)
69	2.5	1.0
70	2.25	1.25
71	2.0	1.5
72	1.75	1.75
73	1.5	2.0
74	1.25	2.25
75	1.0	2.5

Local out of flatness conditions, such as minor scratches, are allowable provided the flatness reading is unaffected.

3.16 Workmanship.

3.16.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.16.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.16.3 Burrs. Parts shall be free from burrs.

3.16.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.16.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.16.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 8 complete assemblies (see dwg 8878819) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table IX. Tests shall be conducted in order specified in Table IX.

Table IX

First article inspection and tests

Inspection or Test	Non-Destr	Destr	Testing Sequence Unit Nos. 1 2 3 4 5 6 7 8
1. Classification of characteristics (see 4.3.2.1)	X		X X X X X X X X
2. *Gas charge pressure and weight (see 3.12)	X		X X X X X X X X
3. *Case leakage (see 3.6)	X		X X X X X X X X
4. Circuit resistance (non-operating) (see 3.7)	X		X X X X X X X X
5. Current carrying capacity (see 3.8)	X		X X X X X X X X
6. Insulation resistance (see 3.9)	X		X X X X X X X X
7. **High temperature (see 3.11.3)	X		X X X
8. **Low temperature (see 3.11.4)	X		X X
9. **Transportation vibration @ 165 ± 5°F (see 3.11.1.1)	X		X X X X X
10. **Transportation vibration @ -65 ± 5°F (see 3.11.1.2)	X		X X X X
11. **Impact shock @ 165 ± 5°F (see 3.11.2.1)		X	X X X
12. **Impact shock @ -65 ± 5°F (see 3.11.2.2)		X	X X
13. **Temperature & Humidity (see 3.11.6)		X	X
14. ***"Go" operation (input rate: 1200°/second) @ 80 ± 20°F (see 3.10.2.1)	X		X

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

		1	2	3	4	5	6	7	8
15. ***"Go" operation (input rate: 500°/second) cycle #2, 165 + 5°F (see 3.10.2.2)	X		X		X		X		X X
16. ***"Go" operation (input rate: 500°/second) cycle #3, @ -40 ± 5°F (see 3.10.2.2)	X			X		X		X	X
17. ***"No-go" operation (low input rate: 380°/second) cycle #1, @ 80 + 20°F (see 3.10.1)	X		X	X	X	X	X	X	X
18. ***"No-go" operation (low input rate: 380°/second) cycle #2, @ 165 + 5°F (see 3.10.1)	X		X	X	X	X	X	X	X
19. ***"No-go" operation (low input rate: 380°/second) cycle #3, @ -40 ± 5°F (see 3.10.1)	X		X	X	X	X	X	X	X

*Not applicable when operated by approved simulation of piston motor frangible seal activation.

**After each environmental test, 7 thru 13, each gyroswitch shall satisfy the requirements of test 2 thru 6.

***After the first functional operation test, the insulation resistance shall be measured prior to removal of the piston motors. Prior to performing the circuit resistance and current carrying capacity tests, the piston motors shall be removed and replaced by a shorting circuit at the terminals to which the piston motors were attached. After any functional operational test the gyroswitch shall meet the requirements of circuit resistance (see 3.7), insulation resistance (see 3.9), and current carrying capacity (see 3.8).
After any functional operation test which is to be followed by additional tests, per Table IX, the gyroswitch shall be recharged to the pressure shown on the Data Plate, 8878844, prior to the start of first article testing. Subsequent functional operation tests (after the first) are to be performed using simulated functioning of the piston motors and frangible seal.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.2.5 For first article testing, after exposure to any environment, the cage switch voltage drop criteria shall be 0.200 volts maximum in lieu of 0.125 volts maximum.

4.3 Quality conformance inspection.

4.3.1 Lot formation. The formation of an inspection lot and related functions shall be in accordance with Section 5, MIL-STD-105. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table X and Table XI), and inspected for all characteristic in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

Table X
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

Table XI
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristice (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1 Rotor assembly (see dwg 8877919).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Excessive unbalance of Rotor Assembly	Gage
	102. Evidence of poor workmanship (see 3.16)	Visual
	103. Magnet protrudes surface	Gage
Minor:	None defined.	

4.3.2.1.2 Gimbal assembly (see 8877941).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Excessive gimbal unbalance	Gage
	102. Absence of sealing compound (rotor shaft hex nut only)	Visual
	103. Incorrect rotor coast time (see 8877919 for rpm)	Gage
	104. Pin not oriented	Visual
	105. Evidence of poor workmanship (see 3.16)	Visual
	106. Incorrect torques	Gage
Minor:	None defined.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.3.2.1.3 Bellows assembly, exhaust (see dwg 8878813).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Excessive leak rate	Gage
102.	Incorrect overall length	Gage
103.	Evidence of poor workmanship (see 3.16)	Visual
Minor:	None defined.	

4.3.2.1.4 Ring assembly, contact (see dwg 8878821).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	0.270 ± 0.007 dimension, incorrect	Gage
102.	Excessive radial misalignment of contacts	Gage
103.	Evidence of poor workmanship (see 3.16)	Visual
104.	Evidence of solder on glass insulators (15)	Visual
105.	Glass insulators (15) cracked, chipped or flaked	Visual
Minor:	None defined.	

4.3.2.1.5 Terminal assembly (see dwg 8877874).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Surface - B - not parallel to - C -	Gage
102.	Absence of adhesive	Visual
103.	Evidence of poor workmanship (see 3.16)	Visual
Minor:	None defined.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.3.2.1.6 Terminal assembly (see dwg 8878826).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Surface - B - not parallel to surface - C -	Gage
102.	Absence of adhesive	Visual
103.	Evidence of poor workmanship (see 3.16)	Visual
Minor:	None defined.	

4.3.2.1.7 Support assembly, gimbal (see dwg 9285346).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper gimbal endplay	Gage
102.	Evidence of poor workmanship (see 3.16)	Visual
Minor:	None defined.	

4.3.2.1.8 Dome subassembly (see dwg 8878743).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Characters missing, incorrect or illegible	Visual
102.	Dimension 2.190 + .005 not to size	Gage
103.	0.138 + 0.004 diameter hole improperly located (15)	Gage
104.	(3) .125 + .005 - .001 diameter holes improperly located	Gage
105.	Evidence of poor workmanship (see 3.16)	Visual
106.	Cracks and porosity in electron beam weld	Visual
Minor:	None defined.	

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

4.3.2.1.9 Dome assembly (see dwg 8878817).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper soldering (see 3.13) of contact plate	Visual
102.	Absence of potting compound, encapsulating shrink sleeving and terminals (as shown on drawing)	Visual
103.	Connector, Electrical (J10) improperly oriented	Visual
104.	Evidence of poor workmanship (see 3.16)	Visual
105.	Insulation improperly installed	Visual
106.	Insulation resistance to case less than 20 megohms	Gage
107.	Improper soldering (see 3.13) of dome wires	Visual
Minor:	None defined.	

4.3.2.1.10 Bellows assembly, cage (see dwg 8878793).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Excessive leak rate	Gage
102.	Evidence of poor workmanship (see 3.16)	Visual
Minor:	None defined.	

4.3.2.1.11 Wafer assembly, cage (see dwg 8878784).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Blades secured to wrong side of Wafer, Switch (12)	Visual
102.	Evidence of poor workmanship (see 3.16)	Visual
103.	Improper soldering (see 3.13)	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

104. Marking incorrect, illegible or missing.....Visual

Minor: None defined.

4.3.2.1.12 Mount assembly, center (see dwg 9285347).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Speed control improperly adjusted.....	Gage
102.	Evidence of obstruction in orifice, sustaining.....	Visual
103.	Excessive leak rate.....	Gage
104.	Evidence of poor workmanship (see 3.16).....	Visual
105.	Improper adjustment of movement of Stop, Limiting.....	Gage
106.	Pin, Spring missing.....	Visual
107.	Absence of sealing compound on Screw, Adjust Only.....	Visual
108.	Insufficient contact pressure (12 contacts).....	Gage
Minor:	None defined.	

4.3.2.1.13 Cap, exhaust (see dwg 8877928).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Diameter - A - not to size.....	Gage
102.	0.375 - 0.002 diameter not to size.....	Gage
103.	0.375 - 0.002 diameter not concentric to diameter - A -.....	Gage
104.	Length of diameter -A- incorrect.....	Gage
105.	0.375 ± 0.002 dimension incorrect.....	Gage
106.	Evidence of poor workmanship (see 3.16).....	Visual
107.	Evidence of radial tool marks on flange.....	Visual
108.	Incorrect machine finish on flange.....	Gage
Minor:	None defined.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.3.2.1.14 M46 Gyroswitch unwelded assembly (see dwg 9285348).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Gimbal stops (2) improperly adjusted	Gage
102.	Improper contact switch angle	Gage
103.	Improper soldering (see 3.13)	Visual
104.	Absence of sealing compound	Visual
105.	Evidence of poor workmanship (see 3.16)	Visual
106.	"O" ring missing or damaged	Visual
107.	Improper contact pressure	Gage
Minor:	None defined.	

4.3.2.1.15 M46 Gyroswitch assembly (see dwg 8878819).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect location of 4 mounting holes (see dwg 8877904)	Gage
102.	Incorrect diameter of 4 mounting holes (see dwg 8877904)	Gage
103.	Improper soldering (see 3.13)	Visual
104.	Loose or missing hinge pin (with cover removed)	Visual/Manual
105.	Absence of sealing compound on exhaust cap	Visual
106.	Absence of sealing compound on plug, final seal	Visual
107.	Incorrect torques	Gage
108.	Weight and charging pressure missing on data plate (see 3.12)	Visual
109.	Evidence of poor workmanship (see 3.16)	Visual
110.	Evidence of cracks and porosity in electron beam weld of base to center mount assembly	Visual
111.	Absence of ring, retaining (cover removed)	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

112. Alignment pin diameter hole in base incorrect (see dwg 8877904).....Gage
113. Incorrect depth of alignment pin hole (see dwg 8877904).....Gage
114. Incorrect location of alignment pin hole (see dwg 8877904).....Gage
115. Inability to mate with mating connector.....Manual
116. Any parts missing, loose, damaged or improperly assembled.....Visual
117. Incorrect diameter of raised portion of base (see dwg 8877904).....Gage
118. Incorrect width to raised portion of base (see dwg 8877904) before welding.....Gage
119. Base not lapped to within correct wavelength after pressurization (see 3.15).....Gage
120. Evidence of cracks and porosity in electron beam weld of center mount assembly to dome.....Visual
121. Absence of protective strip coating in lapped area of base.....Visual
122. Objective evidence relative to material missing or improper (see 3.1).....Visual
123. Serialization missing or improper (see 3.4).....Visual
124. Gas charge pressure incorrect (see 3.12).....Gage
125. Gas charge weight incorrect (see 3.12).....Gage

Minor: None defined.

4.3.2.1.16 Connector assembly (see dwg 8898847).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Improper soldering (see 3.13).....Visual
102. Evidence of poor workmanship (see 3.16).....Visual
103. Any part missing, loose, damaged or incorrectly assembled.....Visual

Minor: None defined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.3.2.1.17 Cap assembly, pre-load (see dwg 8877827).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper soldering (see 3.13)	Visual
102.	Evidence of poor workmanship (see 3.16)	Visual
103.	Leads not soldered to correct terminals (2)	Visual
104.	Any parts loose, damaged, missing or improperly assembled	Visual/Manual
105.	Service leads not covered with epoxy resin (2)	Visual
Minor:	None defined.	

4.3.2.1.18 Contact ring, cage wafer and function switch assembly (see dwg 9285345).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper soldering (see 3.13)	Visual
102.	Evidence of poor workmanship (see 3.16)	Visual
103.	Any part missing, loose, damaged or incorrectly assembled	Visual
Minor:	None defined.	

4.3.2.1.19 Preparation for delivery (see dwg 9274158).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packaging, and packing not in accordance with dwg 9274158	Visual
102.	Marking not in accordance with dwg 9274158	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.16)	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Functional operation	Major
b. Gas charge pressure and weight	Major
c. Case leakage	Major
d. Circuit resistance	Major
e. Current carrying capacity	Major
f. Insulation resistance	Major
g. Temperature shock	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Gas charge pressure and weight, case leakage, circuit resistance, current carrying capacity, insulation resistance, and temperature shock. Each gyroswitch shall be tested in accordance with 4.4.1 thru 4.4.4; 4.4.6.6 and 4.4.7. If any item fails to comply with 3.6 thru 3.9; 3.11.5 and 3.12, the item shall be rejected and removed from the lot.

4.3.3.2 Functional operation. Each gyroswitch shall be subjected to the functional operation tests in accordance with 3.10.1 Cycle #1, and Cycles #1 and #2 in accordance with 3.10.2.2, in that order. The above tests shall all be done prior to installation of the piston motors and frangible seal using the gas charge pressure per 3.12 and an approved simulation of piston motor frangible seal activation. After satisfactory completion of these tests, each gyroswitch shall meet the requirements of insulation resistance, circuit resistance, and current carrying capacity (with a shorting circuit substituted for the piston motors). The gyroswitches shall have test piston motor and insulation assembly (9274282) and the test exhaust cap (8877928) installed and pressurized to the gas charge pressure per 3.12. Each unit shall then be tested to the Function Performance Test in accordance with 3.10.2.2 Cycle #3. The gyroswitches shall have their new piston motor and insulation assembly (9274285) and their new exhaust cap (8877928) installed and pressurized to the pressure per 3.12. These units shall then be tested to assure they meet the requirements of 4.3.3 b to f. If any gyroswitch fails to comply with any of the above requirements, it shall be classified defective and removed from the lot.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-8878819. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. All tests are conducted at room ambient temperature (75 ± 10°F) unless specified otherwise.

4.4.1 Case leakage. The case leakage rate of the gyroswitch shall be measured by an approved mass spectrometer. The test chamber shall be evacuated to one millitorr, or less, and the leak rate shall comply with the leakage rate specified in 3.6.

4.4.2 Circuit resistance. The circuit resistance, except for U to J, shall be measured by the voltammeter method using a current of 100 ma maximum. The resistance between the applicable J10 terminals shall not exceed the ohmic levels specified in 3.7. The current shall be limited to twenty (20) milliamperes maximum for J10 terminals S to N and T to D. The current shall be limited to 0.015 ampere for J10 terminals U to J.

4.4.3 Current carrying capacity. The current carrying capacity shall be measured with an approved electrical tester using 37 vdc maximum. The maximum voltage drop between the applicable J10 terminals shall not exceed the voltage levels specified in 3.8. Current between J10 terminals a to E and b to M shall be 0.5 ± 0.03 amperes, and for terminals T to R and S to C the current shall be 4.75 ± 0.25 amperes for a maximum of 2.0 second.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

4.4.4 Insulation resistance. The insulation resistance shall be measured with an approved electrical tester using 500 vdc ± 50 vdc applied for 60 seconds ± 10 seconds for each test in accordance with 3.9. The current shall be limited to 5 milliamperes.

4.4.5 Functional operation.

4.4.5.1 Operational time. The gyroswitch shall be rigidly mounted on a suitable rate table in an upright position and a constant input rate applied in compliance with the requirements of 3.10, as applicable. Input rates shall be in a clockwise direction as viewed from the connector end and shall be applied prior to initiation of the gyroswitch (See Figure 1).

4.4.5.2 Functional switch voltage drop test. Using a 37 vdc maximum power supply to the specified terminals and an applied current of 0.5 ± 0.03 amperes, the gyroswitch shall comply to the applicable requirements in 3.10.

4.4.5.3 Initiation. An electrical input of 37 vdc maximum and 4.5 ± 0.15 amperes shall be applied between J10 terminals T and D, and S and N. The initiation voltage shall be applied for a minimum of 20 milliseconds (ms). Not applicable when operated by approved simulation of piston motor frangible seal activation.

4.4.6 Environmental testing (see 3.11).

4.4.6.1 Transportation vibration. The gyroswitch shall be securely mounted on a fixture capable of being readily positioned in each of the three (3) perpendicular axes, (see Figure 1). The gyroswitch shall then be mounted on a vibration producing device and tested to the requirements in 3.11.1.

4.4.6.2 High temperature. The gyroswitch shall be placed into a temperature chamber. The gyroswitch shall be maintained at 165 ± 5°F for four hours minimum or until the temperature of the gyroswitch has stabilized (see 6.4). When using simulated piston motors and frangible exhaust seals, the gyroswitch shall be initially pressurized to 1000 psig (at ambient) and then placed into a temperature chamber. The gyroswitch shall be maintained at 165 ± 5°F for 3 hours minimum. Then for one additional hour, the unit shall be pressurized to the charge pressure specified on the data plate plus the additional pressure resulting of a temperature increase to 165°F. This additional pressure shall be maintained externally until Gyroswitch is removed from the chamber.

4.4.6.3 Low temperature. (-40 ± 5°F). The gyroswitch shall be placed into a temperature chamber. The gyroswitch shall be maintained at -40 ± 5°F for four hours minimum or until the temperature of the gyroswitch has stabilized (see 6.4).

4.4.6.4 Low temperature. (-65 ± 5°F). The gyroswitch shall be placed into a temperature chamber. The gyroswitch shall be maintained at -65 ± 5°F for four hours minimum or until the temperature of the gyroswitch has stabilized (see 6.4).

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

4.4.6.5 Impact shock. The gyroswitch shall be securely positioned on a fixture capable of being readily mounted in its three (3) mutually perpendicular axes (see Figure 1). The gyroswitch shall be mounted on a shock producing device and tested to the requirements of 3.11.2.

4.4.6.6 Temperature shock. The gyroswitch shall be placed in a chamber that has been pre-conditioned to -65 ± 5°F and maintained at that temperature. The gyroswitch shall remain at that temperature for a minimum of four hours or until the temperature of the gyroswitch has stabilized (see 6.4). At the end of that time the gyroswitch shall be transferred to a separate chamber within five minutes which has been preconditioned to 165 ± 5°F. The gyroswitch shall be maintained at the temperature for four hours and then transferred within five minutes to the first chamber at -65°F completing one cycle. The gyroswitch shall be subjected to a total of three cycles as defined above and then removed from the test chamber and allowed to return to ambient conditions.

4.4.6.7 Temperature and humidity. The gyroswitch shall be subjected to a temperature-humidity cycling test in accordance with MIL-STD-331, Test 105. This test shall be run only for the first 7 days of the 28 day JAN cycle.

4.4.7 Gas charge pressure and weight. The gas charge pressure in psig shall be measured at the gyroswitch filling port with equipment capable of measuring pressure within 1% of gyroswitch operational pressure. This measured pressure shall be recorded.

The temperature of the compressed gas, gas pressuring system and gyroswitch shall be stabilized (see 6.4) prior to pressurization. Temperature of the gyroswitch, immediately prior to pressurization, shall be measured with equipment capable of indicating temperatures within 1/2°F. This temperature shall be recorded.

Gas pressurization rate at the gyroswitch shall not exceed 500 psi/minute.

The weight of the gyroswitch shall be determined and recorded before and after pressurization using an approved scale capable of weighing within 0.050 gram.

Test codes (see 3.5) shall be as follows:

Gas charge pressure: 031301

Temperature at which gas charge pressure is measured: 031302

Gyroswitch weight before pressurization: 031303

Gyroswitch weight after pressurization (final unit weight): 031304

Gas charge weight: 031305

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

4.4.8 Flatness. The gyroswitch shall be temperature-conditioned at 72 + 3°F for a minimum of four hours. The base flatness shall be measured over a concentric area forming 80% of the diameter of the raised portion using an approved optical flat and light source. The measurement shall be accomplished within the temperature range from 69 to 75°F of the gyroswitch. Monochromatic yellow (wave length of 5876 Angstrom units) shall be used as a light source. Image of light wave pattern shall be photographically recorded. Optical flatness shall comply with 3.15. Test code for optical flatness shall be 031601 and test code for temperature 031602.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The gyroswitch shall be packaged in accordance with dwg 9274158 using the specified moisture barrier.

5.1.2 Level C. The gyroswitch shall be packaged in accordance with dwg 9274158 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The gyroswitch shall be packed in accordance with dwg 9274158.

5.3 Marking. Marking shall be as specified in accordance with dwg 9274158.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5). Data specified in 4.3.4.1 and 3.5 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.

6.4 Temperature stabilization. The temperature is considered stabilized when three consecutive temperature readings, taken at 5 minute intervals, are within ± 1°F of each other.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2642B, dated 22 August 1972, and Amendment 3, dated 20 May 1975.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A228

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

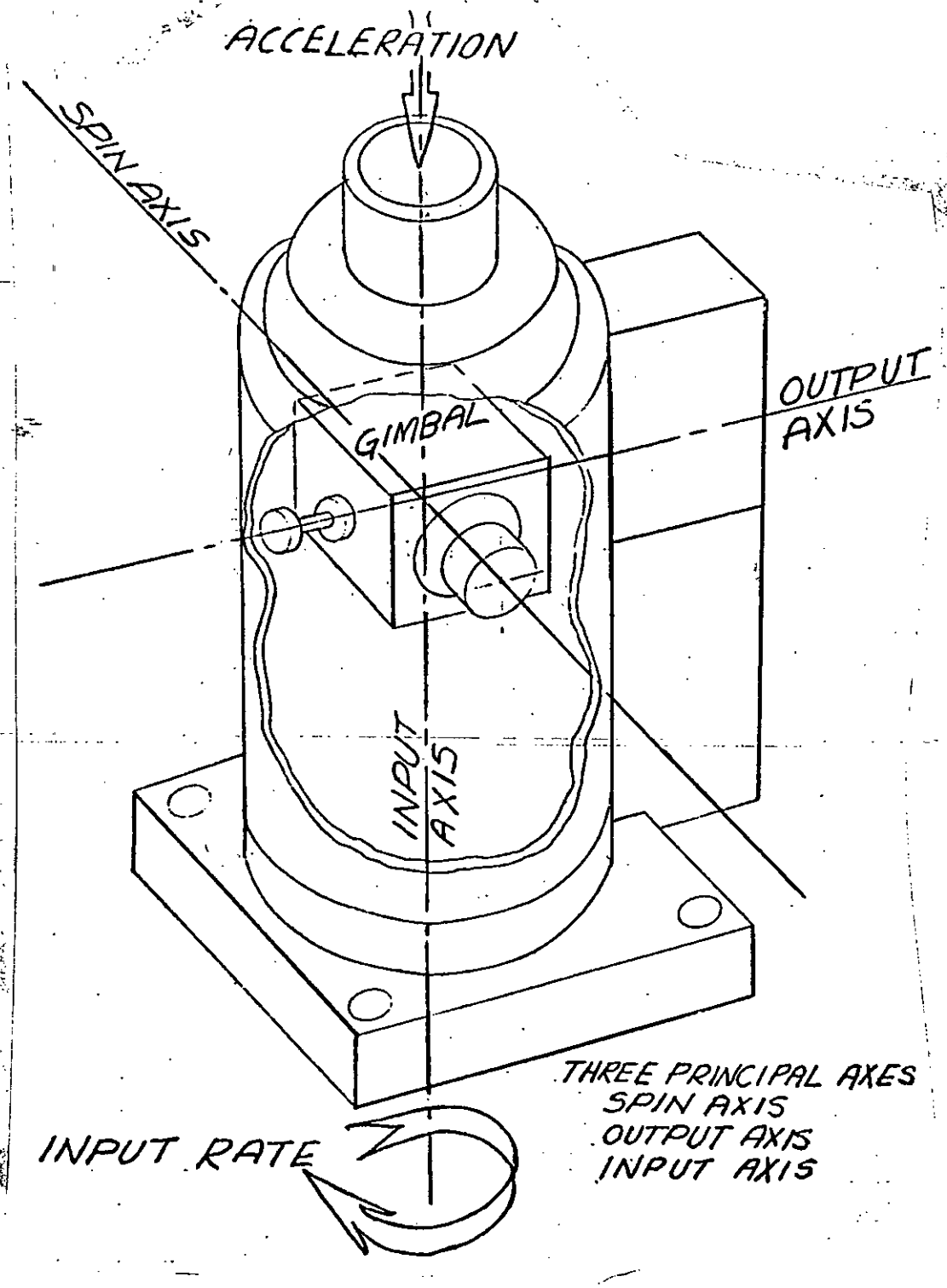


Figure 1

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:56Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 **1426**

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:56Z
Check the source to verify that this is the current version before use.